

Glossary



Notation	Description
<i>pp</i> -chain	proton-proton chain.
<i>pp</i> 1-chain	path 1 proton-proton chain.
<i>pp</i> 2-chain	path 2 proton-proton chain.
<i>pp</i> 3-chain	path 3 proton-proton chain.
AGB	Asymptotic Giant Branch.
BBH	Binary Black Hole.
BH	Black Hole.
BH-WD	Black Hole - White Dwarf Binary.
BNS	Binary Neutron Star.
BPASS	Binary Population and Spectral Synthesis.
CCSN	core-collapse supernovae.
CDF	Cumulative Distribution Function.
CEE	Common Envelope Evolution.
CO	Compact Object.
CO core	Carbon-Oxygen core.
CPTA	Chinese Pulsar Timing Array.
DTD	Delay-Time Distribution.
EOS	Equation of State.

Notation	Description
EPTA	European Pulsar Timing Array.
GR	General Relativity.
HRD	Hertzsprung-Russell Diagram.
IMF	Initial Mass Function.
ISCO	innermost stable circular orbit.
ISM	Interstellar Medium.
JWST	James Webb Space Telescope.
KS	Kolmogorov-Smirnov.
LIGO	Laser Interferometer Gravitational Wave Observatory.
LISA	Laser Interferometer Space Antenna.
LMC	Large Magellanic Cloud.
LVK Collaboration	LIGO-Virgo-KAGRA Collaboration.
MLT	Mixing Length Theory.
MS	Main Sequence.
MWEG	Milky-Way Equivalent Galaxy.
NANOGrav	North American Nanohertz Observatory for Gravitational Waves.
NS	Neutron Star.
NS-BH	Neutron Star - Black Hole Binary.
NS-WD	Neutron Star - White Dwarf Binary.
O3	Operating Run 3.
O4	Operating Run 4.
ONe core	Oxygen-Neon core.

Notation	Description
PDF	Probability Density Function.
PPTA	Parkes Pulsar Timing Array.
RGB	Red Giant Branch.
RLOF	Roche Lobe overflow.
SFH	Stellar Formation History.
SFRD	Star Formation Rate Density.
SMC	Small Magellanic Cloud.
SPH	Smoothed-particle hydrodynamics.
TŻO	Thorne-Żytkow Object.
TAMS	Terminal-Age Main Sequence.
USSN	ultra-stripped supernova.
WD	White Dwarf.
WR	Wolf Rayet.
ZAMS	Zero-Age Main Sequence.

Co-Authorship Form

Email:

This form is to accompany the submission of any PhD that contains published or unpublished co-authored work. Please include one copy of this form for each co-authored work. Completed forms should be included in all copies of your thesis submitted for examination and library deposit (including digital deposit), following your thesis Acknowledgements. Co-authored works may be included in a thesis if the candidate has written all or the majority of the text and had their contribution confirmed by all co-authors as not less than 65%.

Please indicate the chapter/section/pages of this thesis that are extracted from a co-authored work and give the title and publication details or details of submission of the co-authored work.

Parts of Chapter 3 are based on “The variable evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material”, submitted to MNRAS and available on the NASA ADS here:

<https://ui.adsabs.harvard.edu/abs/2024arXiv241103000R/abstract>

Nature of contribution by PhD candidate	Performed data analysis, wrote majority of text	
Extent of contribution by PhD candidate (%)	70	

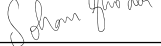
CO-AUTHORS

Name	Nature of Contribution
Jan Eldridge	Wrote part of text, criticism, commentary, and review
Max Briel	Criticism, commentary, and review
Sohan Ghodla	Criticism, commentary, and review

Certification by Co-Authors

The undersigned hereby certify that:

- ☐ the above statement correctly reflects the nature and extent of the PhD candidate’s contribution to this work, and the nature of the contribution of each of the co-authors; and
- ☐ that the candidate wrote all or the majority of the text.

Name	Signature	Date
Jan Eldridge		11/11/2024
Max Briel		19/11/2024
Sohan Ghodla		11/11/2024

Co-Authorship Form

Email:

This form is to accompany the submission of any PhD that contains published or unpublished co-authored work. Please include one copy of this form for each co-authored work. Completed forms should be included in all copies of your thesis submitted for examination and library deposit (including digital deposit), following your thesis Acknowledgements. Co-authored works may be included in a thesis if the candidate has written all or the majority of the text and had their contribution confirmed by all co-authors as not less than 65%.

Please indicate the chapter/section/pages of this thesis that are extracted from a co-authored work and give the title and publication details or details of submission of the co-authored work.

Chapter 4 is based on work from "New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations", published in the Monthly Notices of the Royal Astronomical Society (MNRAS), here: [dx.doi.org/10.1093/mnras/stad977](https://doi.org/10.1093/mnras/stad977)

Nature of contribution by PhD candidate	Wrote majority of the paper, developed most parts of the testing pipeline, later data generation, performed the statistical tests.	
Extent of contribution by PhD candidate (%)	90	

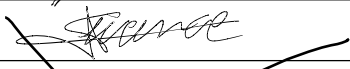
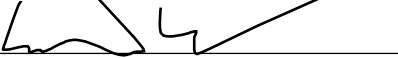
CO-AUTHORS

Name	Nature of Contribution
Jan Eldridge	Original data generation, guidance on project, wrote clarification on the difference between BPASS primary and secondary models, criticism on manuscript
Max Briel	Developing ideas, code commentary, criticism on manuscript
Héloïse Stevance	Developing the test for Ultra-Stripped Supernovae, guidance on project, criticism on manuscript
Reinhold Willcox	Providing single-star pulsar dataset from Willcox+ 2021, criticism on manuscript

Certification by Co-Authors

The undersigned hereby certify that:

- ☐ the above statement correctly reflects the nature and extent of the PhD candidate's contribution to this work, and the nature of the contribution of each of the co-authors; and
- ☐ that the candidate wrote all or the majority of the text.

Name	Signature	Date
Jan Eldridge		11/11/2024
Max Briel		19/11/2024
Héloïse Stevance		21/11/2024
Reinhold Willcox		21/11/2024

